17-01-2026

SHREE

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Fundamentals :-

AI - Artificial Intelligence

Software - website, mobile app, backend, ML / DL
NLP (Natural Language Processing)

SAAS (Software as a Service)
a cloud-based model where you subscribe to use software over the internet, rather than buying and installing it.

Software - whatsapp / zoom, Todo

High stack software - Hospital, stock market, bank, Air booking

"Whatever code you are shipping you are responsiple for that."

4-5 months - Don't use to much AI
(at least 3)

Synthetic data
It is artificially generated data that mimics the patterns, structures, and statistical properties of real-world data but contains no actual personal or sensitive information

Article portions

Problem ⇒ Solution ⇒ Implementation / Conclution

Internet ⇒ 1980s
Web ⇒ 1990s

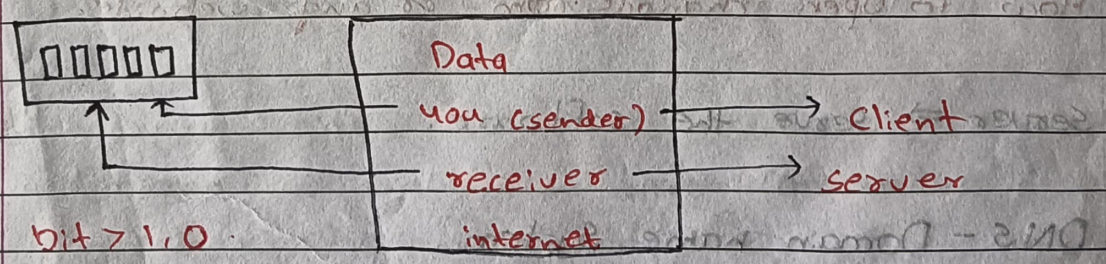
Computers + wires + rules = Network.
⇓
Protocol

Internet :- It is a global communication system that links together thousands of individual networks.
(conection)

Example of protocol :- HTTP / HTTPS , TCP/IP ,
SMTP (sending email) , DNS
(to translate names to IP addresse)

NS2 / NS3
Browser } search
band width

Text / image / video ⇒ Data / Payload



bit > 1, 0
(Data Packets)

Ghar ka address ⇒ IP address
room no ⇒ PORT

= Computer has 65535 port.

one mashin ko multiple port pe multipl software use kar sakte hai

IP - Internet Protocol ⇒ 192.168.1.1
- some IP are reserved you can't buy them, example, apple, google

ISP - Internet Service Provider.

Hardware convert into software.
example
Router (router the data one form to another)

modem (device used to connect internet)

Firewall ⇒ Search
Load balancer
Synthetic data problem

Browser
How to open chaicode.com or any webside.

Server ⇒ Serve the software.

DNS - Domain name System
- DNS turns domain names into IP addresses.

Decentralization
Delegation
Loose coupling
} Search

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DNS is poisoning

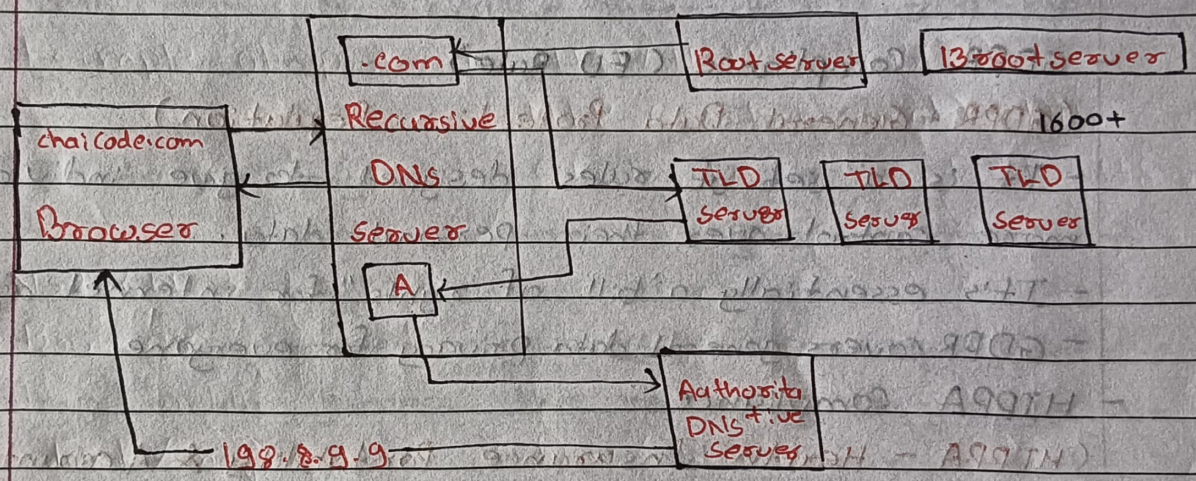
- DNS (Who should I talk to next)

(name of DNS server)

Root Server ⇒ 13 root server in DNS
more than 1600 globally it's always change

- It's give you information of TLD (Top level Domain)

Authoratative DNS server (Big Company)
ex: vercel, godaby, netlyfy, hostinger.



DNS Hierarchy } search
Cashing

Commands
ping -c2 google.com
nslookup google.com
dig google.com NS
dig com NS
dig . NS
dig google.com

Linux commands
ipconfig
ifconfig
top command
Ps command
curl ascii.live/forrest
curl wttr.in

Record Types:- A record, CNAME, MX, NS